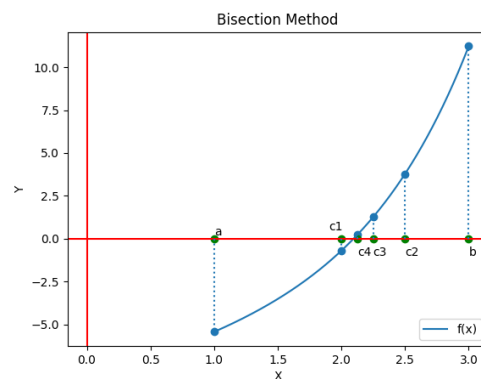


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Lab Status →	Incomplete: ☐	Completed: ☐
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Lab 1: TO FIND A REAL ROOT OF A NON-LINEAR EQUATION BY BISECTION METHOD USING PYTHON PROGRAMMING LANGUAGE

1 Theory

Bisection method is one of the simplest method to solve the non-linear equations. It uses bracketing method to trap the solution between two initial approximations a and b . It uses Intermediate Value Theorem to test whether a root lies within the interval $[a, b]$. Then the interval is then bisected by the point c into the intervals $[a, c]$ and $[c, b]$ and the one interval which doesn't contain the root is discarded. This keeps on going until the root is found with a given error. Here we approximate the root by the mid-points of each intervals in each iterations. The sequence of mid-points converges to the true solution. This method is guaranteed to convergence but slow. It is linearly convergent.



2 Algorithm Bisection

- (i) Start.
- (ii) Define the non-linear equation $f(x)$ and error tolerance E .
- (iii) Take two initial approximations $x = a$ and $x = b$.
- (iv) If $f(a) \times f(b) > 0$, choose other approximations.
- (v) Else compute the mid-point $c = \frac{a+b}{2}$.
- (vi) If $f(a) \times f(c) < 0$, set $b = c$
- (vii) Else set $a = c$.
- (viii) If $|a - b| < E$, then root = $\frac{a+b}{2}$, print root. Go to step (x).
- (ix) Else go to step (iv).
- (x) Stop.

3 Pseudo-code

```
PRINT the title of the experiment
IMPORT necessary libraries
BEGIN IMPORT matplotlib.pyplot AS plt
IMPORT pandas AS pd
IMPORT numpy AS np

#Get equation from user
DISPLAY "Enter the equation in terms of x using python syntax: "
INPUT eqn

#Define function  $f(x)$  using the input equation
FUNCTION  $f(x)$ 
    TRY
        RETURN eval(eqn)
    CATCH SyntaxError, NameError, TypeError AS e
        DISPLAY "Invalid syntax: e"
        RETURN NULL
    END TRY
END FUNCTION

# Get initial guesses
DISPLAY "Enter two initial guesses: "
INPUT a AS FLOAT
INPUT b AS FLOAT

# Check if root exists between  $a$  and  $b$ 
IF  $f(a) * f(b) > 0$  THEN
    DISPLAY "No roots lies between  $a$  and  $b$ "
    EXIT
ELSE
    # Get tolerance and max iterations
    DISPLAY "Enter tolerable error: "
    INPUT E AS FLOAT
    DISPLAY "Enter maximum number of iterations: "
    INPUT N AS INTEGER

    # Initialize variables
    A = [] // Array to store root approximations
    B = [] // Array to store iteration data
    i = 1 // Iteration counter

    #Bisection Method Iterations
    WHILE  $i \leq N$  DO
         $c = (a + b) / 2$  // Calculate midpoint
        Calculate error= $abs(a - b)$ 
        ADD c TO A // Store approximation
        ADD  $[i, a, b, c, f(a), f(b), f(c), \text{ABSOLUTE VALUE}(a - b)]$  to B

    # Update interval
    IF  $f(a) \times f(c) < 0$  THEN
         $b = c$ 
    ELSE
         $a = c$ 
    END IF

    # Check convergence
```

```

        IF error < E THEN
            BREAK
        ELSE

             $i = i + 1$ 
        END IF
    END WHILE

# Check convergence status
    IF  $i > N$  THEN
        DISPLAY "Solution does not converge in " + N + " iterations"
    ELSE
# Convert iteration data to formatted table
        table = CREATE DataFrame FROM B
        WITH COLUMNS ['No. of iterations', 'a', 'b', 'c', 'f(a)', 'f(b)', 'f(c)', 'Error']
        DISPLAY table AS STRING WITHOUT INDEX

        root = c
        DISPLAY "The approximate solution is " + root + " in " + i + " iterations."
    END IF

# Visualization
    x = GENERATE_LINEAR_SPACE(-5, 5, 1000)
    y = CREATE_ZEROS_ARRAY(SAME_SIZE_AS(A))
    A = CONVERT_TO_ARRAY(A)

#Plot the function
    PLOT  $x$  VS  $f(x)$ 
    ENABLE GRID
    DRAW_HORIZONTAL_LINE(0, COLOR = 'red')
    DRAW_VERTICAL_LINE(0, COLOR = 'red')

#Plot root approximations
    SCATTER_PLOT(A, f(A))

# Label iteration numbers
    FOR  $j = 0$  TO LENGTH(A) - 1 DO
        PLACE_TEXT_AT(A[j], 0, TEXT =  $j + 1$ )
    END FOR

#Mark x-intercepts
    SCATTER_PLOT(A, y, MARKER = 'x')

    DISPLAY PLOT
    END

```

4 Code:

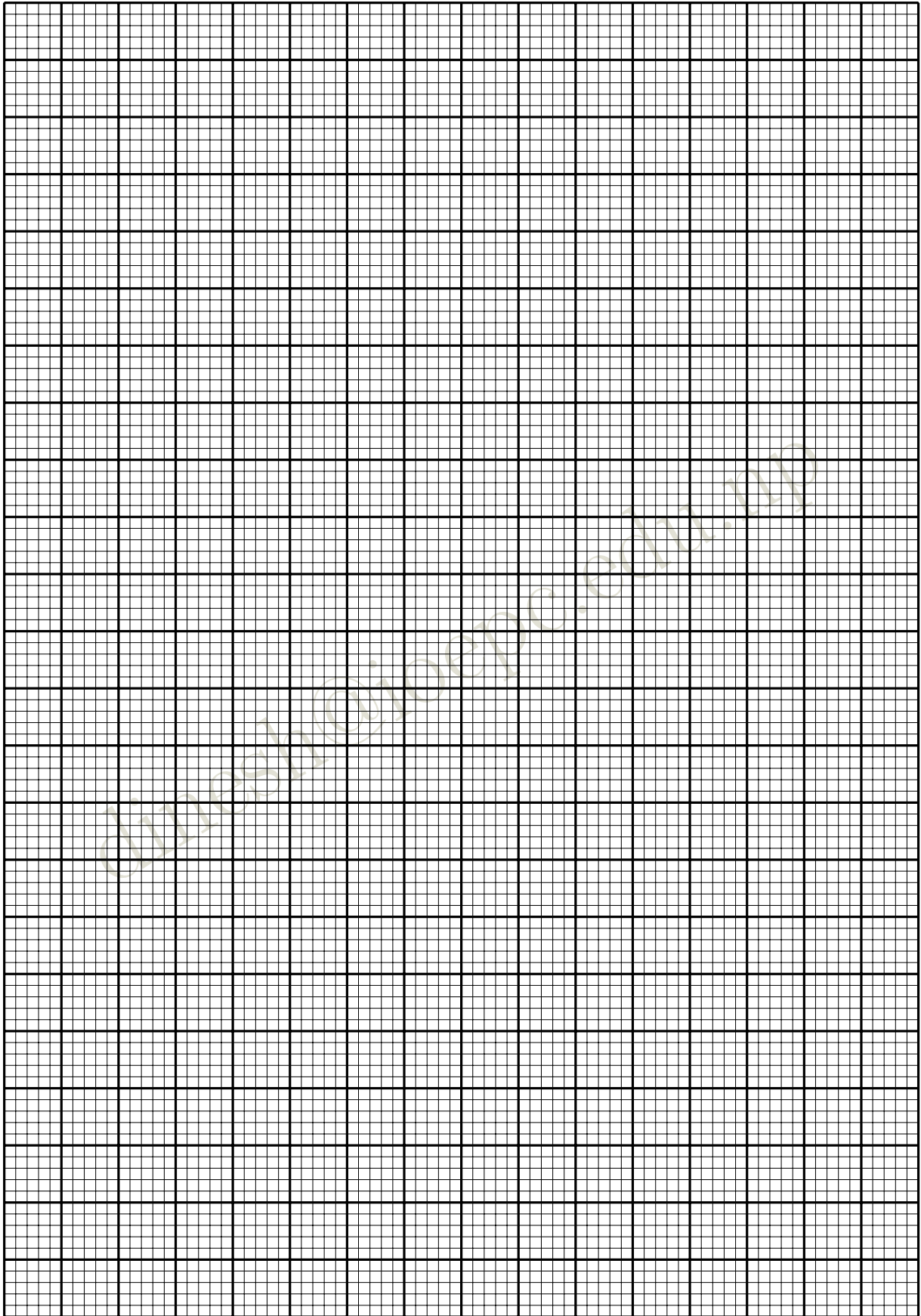


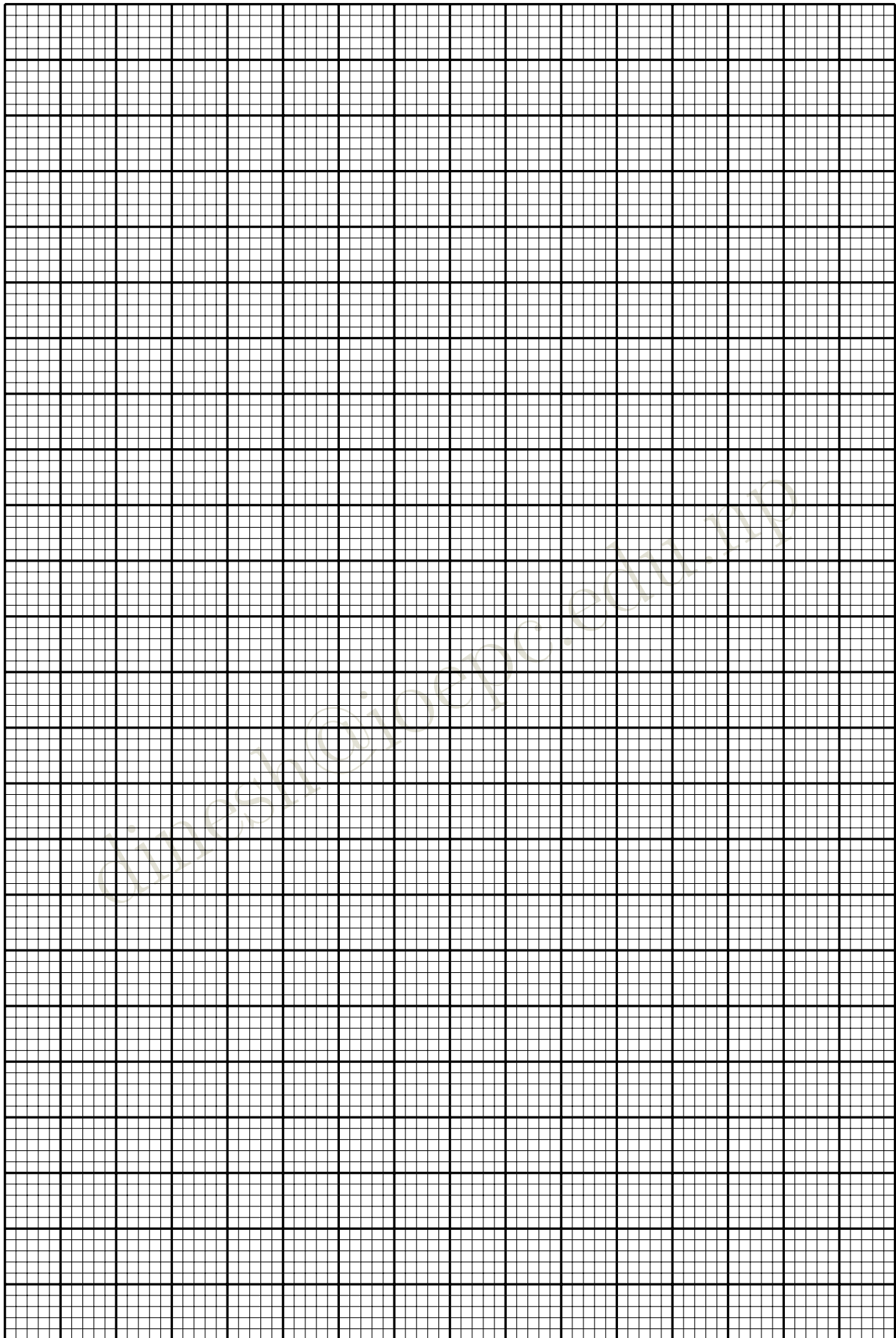
A large grid of dotted lines for writing code. The grid is 20 columns wide and 20 rows high. A diagonal watermark text "dinesh@ioepc.edu.np" is visible across the grid.

5 Inputs and observations

S.N.	Inputs					Output		
	Equations	Initial Guesses		Error	Max. Iterations	Convergence Status	Approx. Solution	No. of iteration
1.	$e^x - x - 2 = 0$	1	2	1e-10	100			
		-2	1	1e-10	100			
		-6	6	1e-10	100			
		-6	-3	1e-10	100			
		-6	8	1e-10	100			
2.	$\cos x - \log x = 0$	1.2	1.4	1e-10	100			
		0	4	1e-10	100			
		-1	5	1e-10	100			
		-20	20	1e-10	100			
		-5	0	1e-10	100			
3.	$x - e^{-x} = 0$	0	-1	1e-10	100			
		0.5	0.6	1e-10	100			
		-1	1	1e-10	100			
		-15	15	1e-10	100			
		-10	10	1e-10	100			
4.	$e^x + \sin(x) - 9 = 0$	1	2	1e-10	100			
		2	2.1	1e-10	100			
		0	4	1e-10	100			
		-5	10	1e-10	100			
		-20	20	1e-10	100			

6 Graph:





7 Findings

(i)

(ii)

(iii)

(iv)

(v)

8 Conclusion: